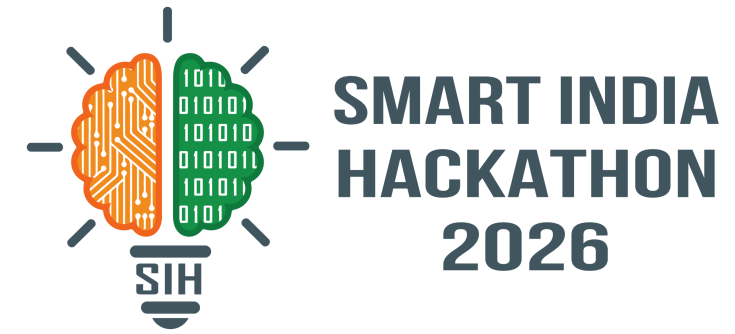


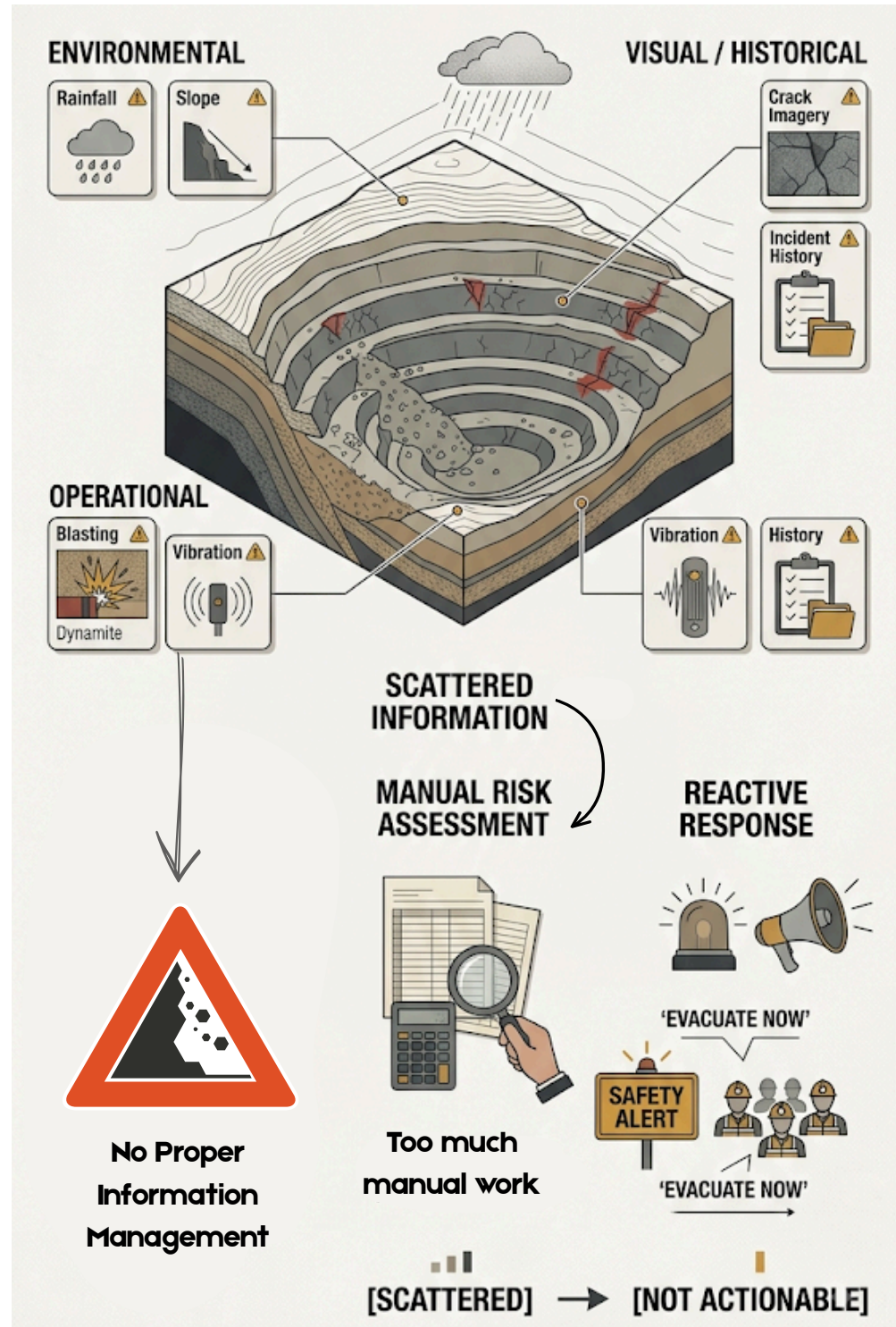
SMART INDIA HACKATHON 2026



- **Problem Statement ID –**
- **Problem Statement Title- Risk-Aware**
- **Decision Support for Open-Pit Mine Safety**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Sangyan**



Problem Faced



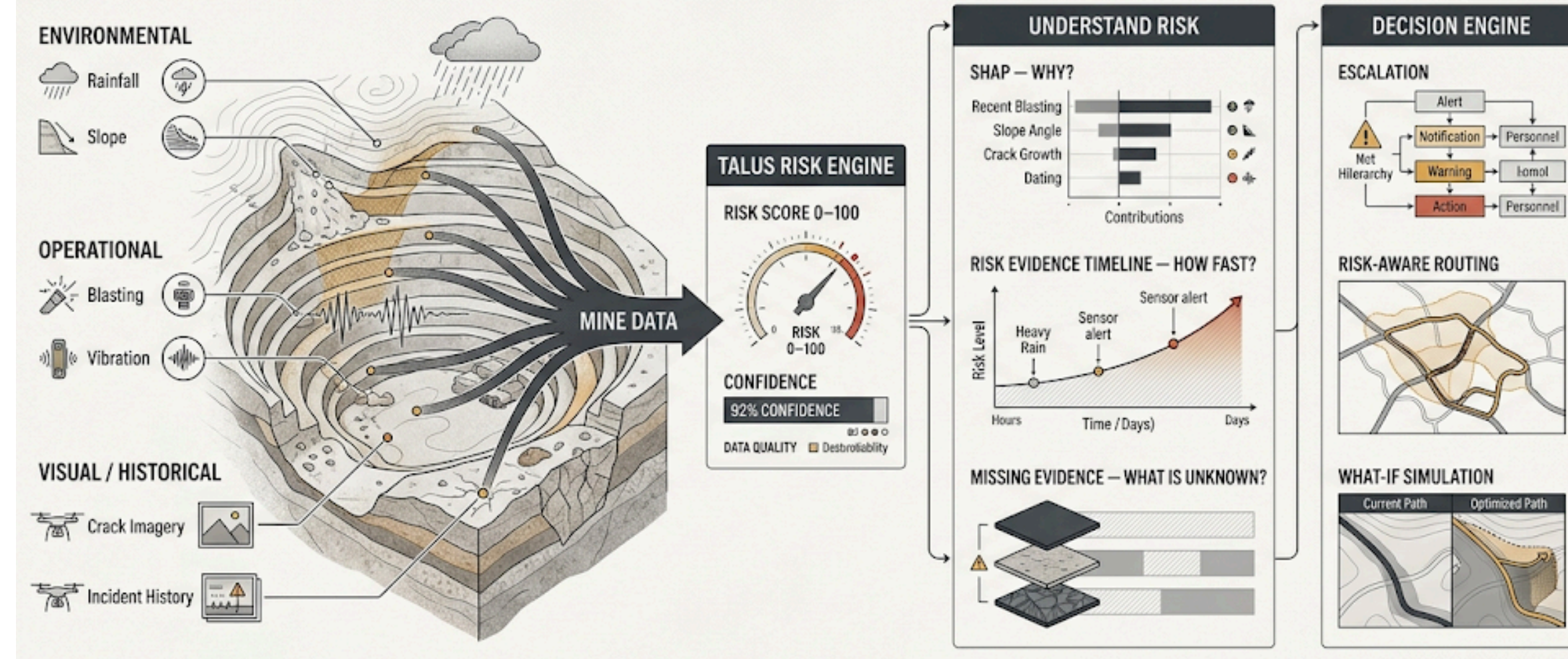
No unified view → delayed understanding → generic response

From scattered signals to
actionable safety decisions.

Our Solution: Talus

Talus converts scattered mine data into explainable risk and actionable safety decisions.

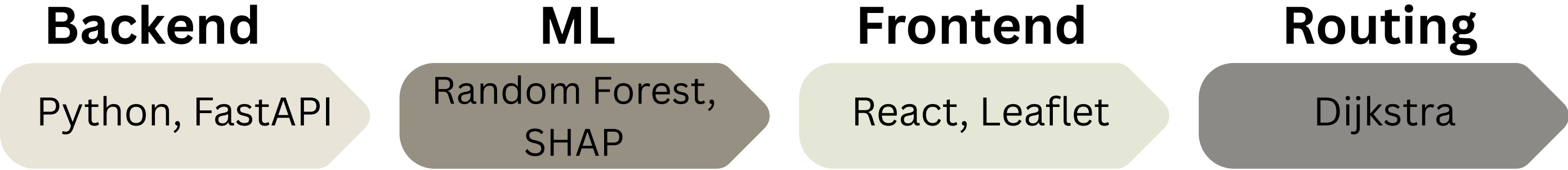
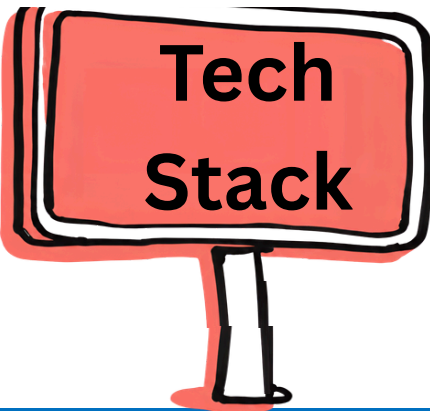
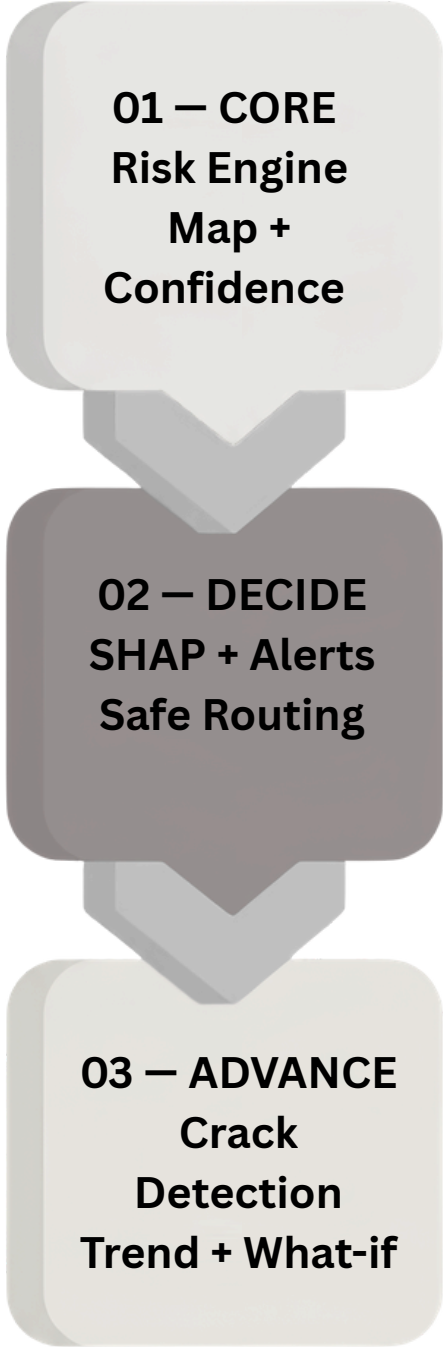
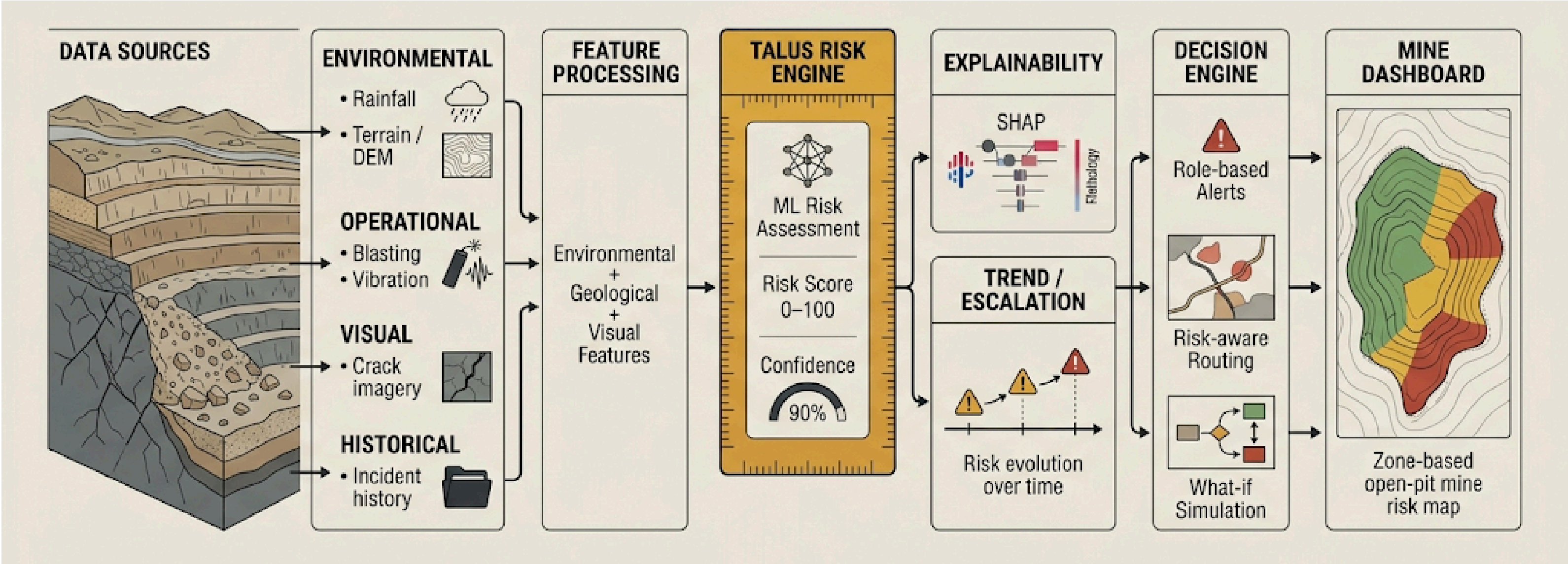
TALUS Risk-Aware Decision Support



TECHNICAL APPROACH

Multi-source data → Risk intelligence → Decision support

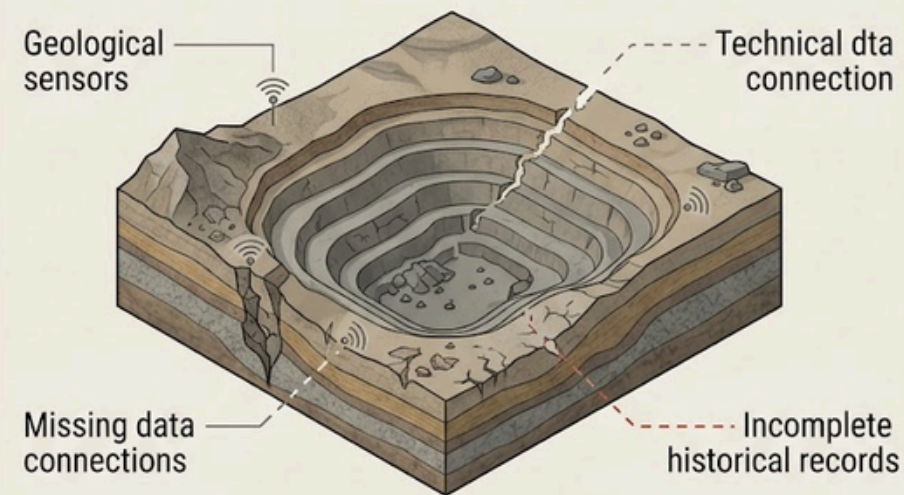
Progressive build:
Core → Decision →
Advanced



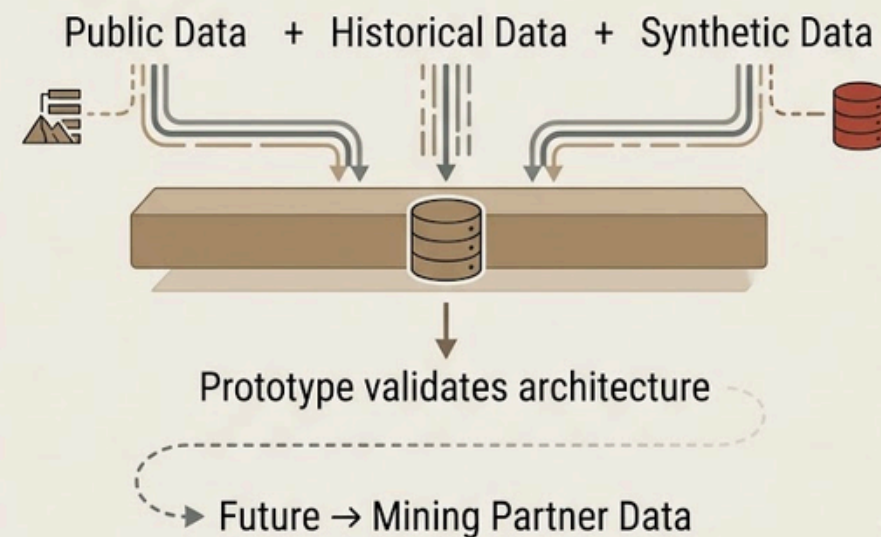
01 – DATA AVAILABILITY

DATA GAP – FEASIBILITY CHALLENGE

PROBLEM: No public Indian mine sensor / incident dataset

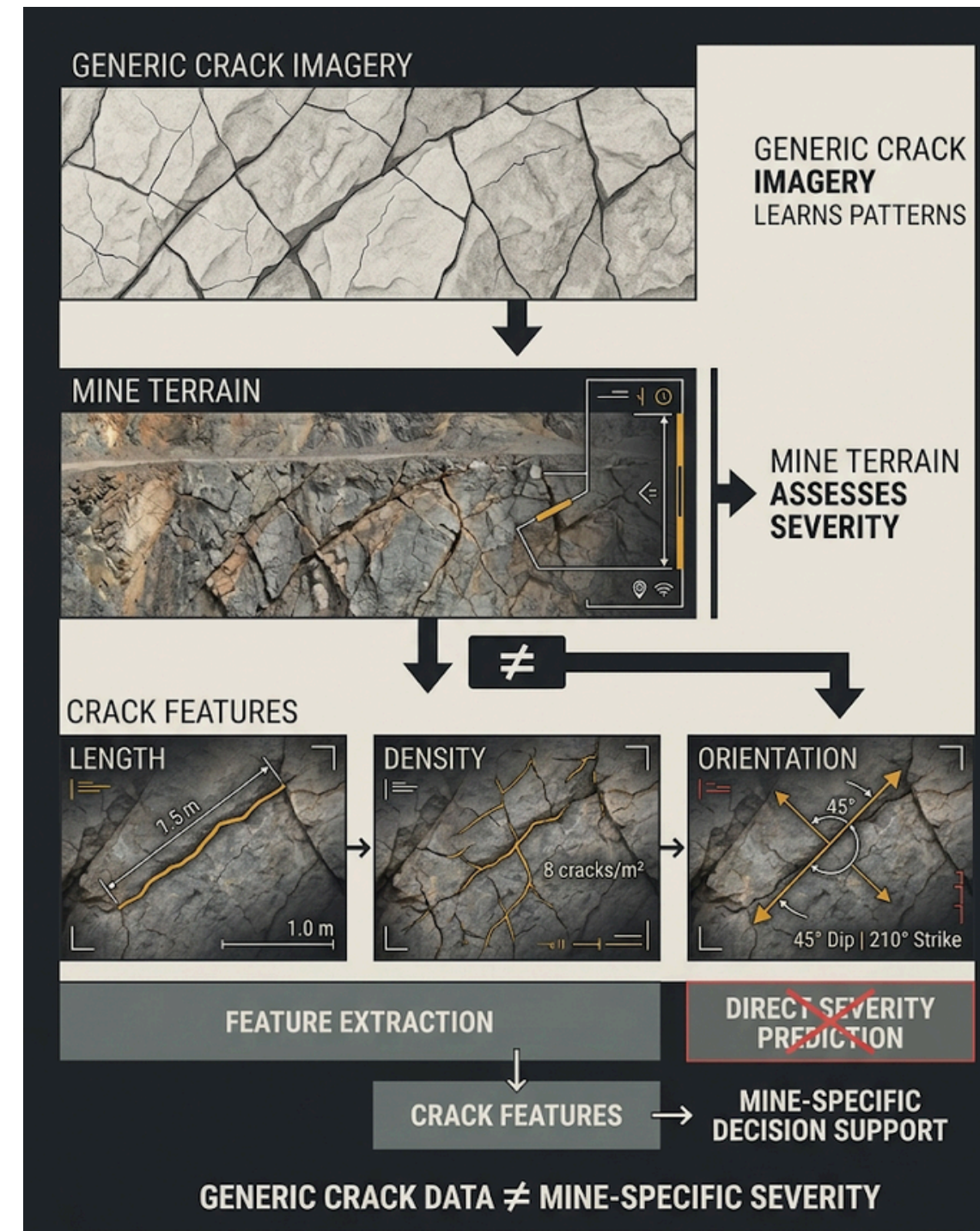


MITIGATION



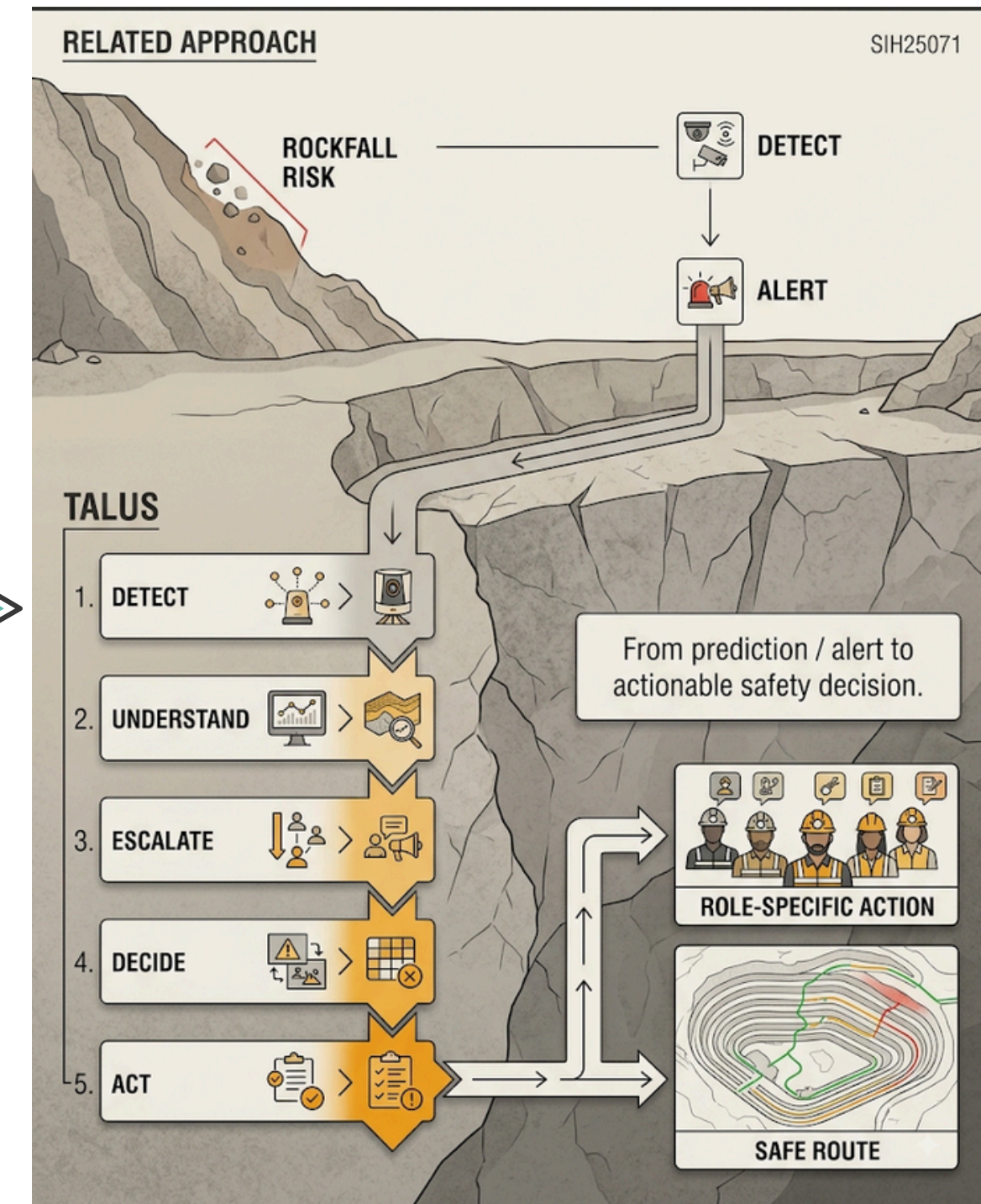
Prototype first; real mine telemetry later.

02 – COMPUTER VISION



Extract measurable crack features, don't claim direct rockfall severity.

03 – DIFFERENTIATION



From “What is the risk?” → “What should we do now?”

IMPACT AND BENEFITS

One risk event → different actions → safer mine

WORKER

- Safe Route Guidance
- Avoid unsafe routes
- Follow safest path

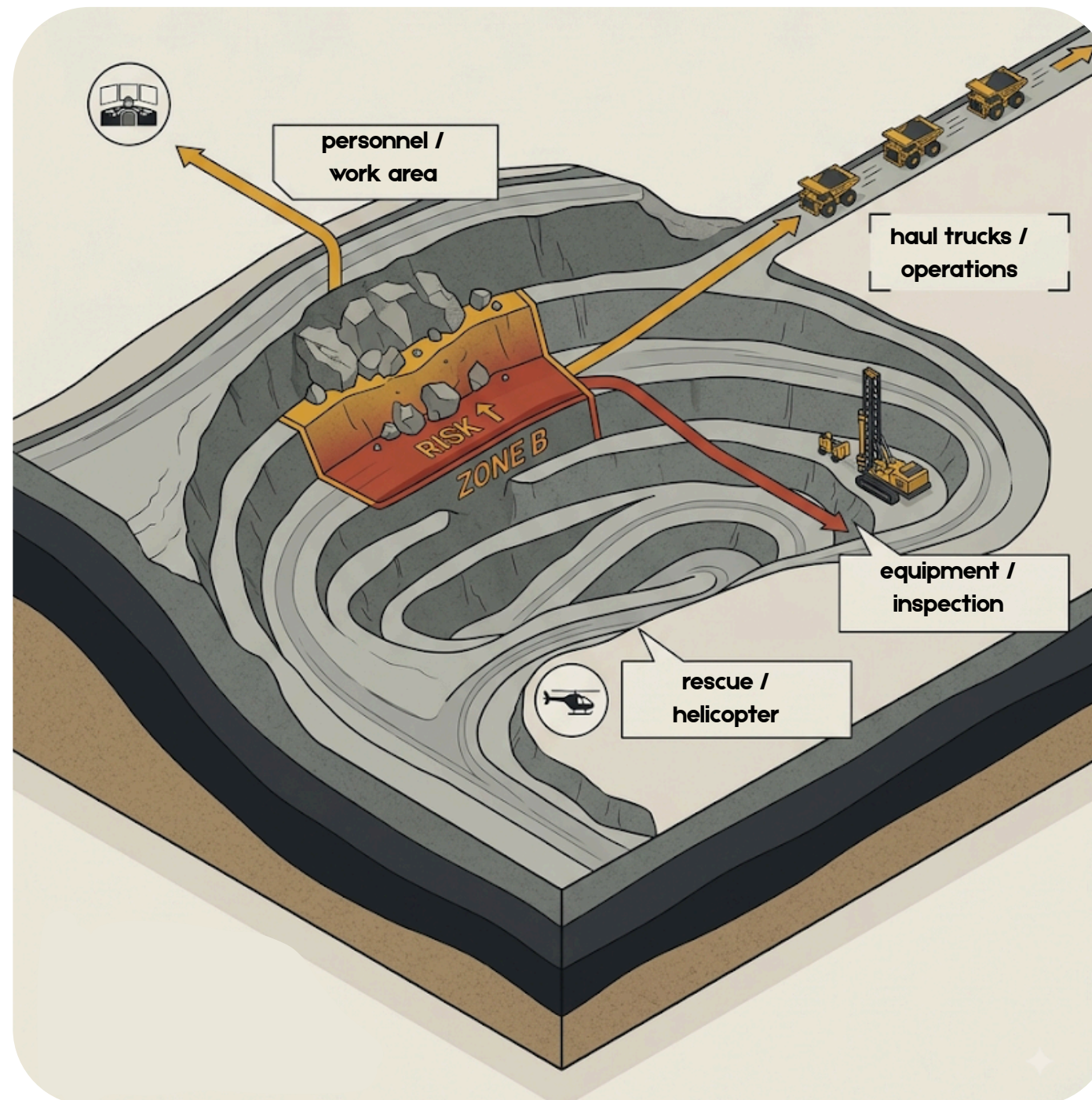
Immediate action

SAFETY OFFICER

- Early Risk Intervention
- Monitor rising zone risk
- Prioritize inspection

Monitor → Escalate

ROLE-SPECIFIC ACTIONS



MINE MANAGER

- Operational Decisions
- Identify people at risk
- Coordinate evacuation

Risk → Decision

RESCUE TEAM

- Risk-Aware Response
- Choose safer approach route
- Avoid unstable zones

Safe Access

SYSTEM-LEVEL BENEFITS

SOCIAL

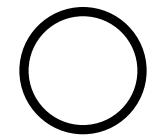
Safer Workforce
Faster risk communication

ECONOMIC

Fewer Reactive Losses
Reduce avoidable disruption

OPERATIONAL

Risk-Based Operations
Prioritized inspections



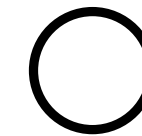
Research & Technical Basis



[1] G. Dharshini, D. Deepika, and C. P., “AI-Based Rockfall Prediction and Alert System for Open-Pit Mines,” in 2025 1st Int. Conf. Advancement in Futuristic Technologies (ICAFT), 2025, doi: 10.1109/ICAFT66710.2025.11452992.

[2] I. P. Senanayake, P. Hartmann, A. Giacomini, J. Huang, and K. Thoeni, “Prediction of rockfall hazard in open pit mines using a regression-based machine learning model,” Int. J. Rock Mech. Mining Sci., vol. 177, p. 105727, 2024, doi: 10.1016/j.ijrmms.2024.105727.

[3] F. Liu, Z. Yang, W. Deng, T. Yang, J. Zhou, Q. Yu, and Y. Mao, “Rock landslide early warning system combining slope stability analysis, two-stage monitoring, and case-based reasoning: A case study,” Bull. Eng. Geol. Environ., vol. 80, no. 11, pp. 8433–8451, 2021, doi: 10.1007/s10064-021-02461-6.



Official / Data Sources



[4] Directorate General of Mines Safety, “The Coal Mines Regulations, 2017,” Government of India. [Online]. Available: [DGMS](#).

[5] India Meteorological Department, “Gridded Rainfall Data.” [Online]. Available: [IMD](#).

[6] ISRO, “Bhuvan / National Database for Emergency Management (NDEM).” [Online]. Available: [ISRO Bhuvan](#).

[7] Smart India Hackathon, “SIH 2026 Themes.” [Online]. Available: [Smart India Hackathon](#).

[8] Ministry of Mines, Government of India, “AI-Based Rockfall Prediction and Alert System for Open-Pit Mines,” Smart India Hackathon Problem Statement SIH25071.

DATA NOTE : No public Indian mine sensor/incident dataset was identified. Prototype validation therefore uses public, historical and synthetic data, informed by the referenced research.